

PVsyst - Simulation report

Grid-Connected System

Project: Madrid_Fixed_Angle_PV_Syst

Variant: Madrid_150MW - Simulation for year no 25 - Simulation for year no 25 - Simulation

Unlimited sheds

System power: 132.7 MWp

Rosas - Spain

Author

University of Edinburgh (United kingdom)



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Project summary

Geographical Site

Rosas
Spain

Situation

Latitude 40.42 °N
Longitude -3.59 °W
Altitude 664 m
Time zone UTC+1

Project settings

Albedo 0.20

Weather data

Rosas
PVGIS api TMY

System summary

Grid-Connected System

Simulation for year no 25

PV Field Orientation

Sheds
Tilt 28 °
Azimuth 0 °

Unlimited sheds

Near Shadings

Mutual shadings of sheds

User's needs

Unlimited load (grid)

System information

PV Array

Nb. of modules 228781 units
Pnom total 132.7 MWp

Inverters

Nb. of units 17 units
Pnom total 116.9 MWac
Pnom ratio 1.136

Results summary

Produced Energy 199298008 kWh/year Specific production 1502 kWh/kWp/year Perf. Ratio PR 71.46 %

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General parameters

Grid-Connected System

PV Field Orientation

Orientation

Sheds	
Tilt	28 °
Azimuth	0 °

Horizon

Average Height	1.2 °
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Unlimited sheds

Sheds configuration

Nb. of sheds	5 units
Unlimited sheds	

Sizes

Sheds spacing	6.60 m
Collector width	3.00 m
Ground Cov. Ratio (GCR)	45.5 %
Top inactive band	0.02 m
Bottom inactive band	0.02 m

Shading limit angle

Limit profile angle	19.8 °
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Models used

Transposition	Perez
Diffuse	Imported
Circumsolar	separate

Near Shadings

Mutual shadings of sheds

User's needs

Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer	Generic
Model	NB-JD580

(Original PVsyst database)

Unit Nom. Power	580 Wp
Number of PV modules	228781 units
Nominal (STC)	132.7 MWp
Modules	9947 string x 23 In series
At operating cond. (50°C)	
Pmpp	122.9 MWp
U mpp	923 V
I mpp	133056 A

Total PV power

Nominal (STC)	132693 kWp
Total	228781 modules
Module area	590999 m²
Cell area	545626 m²

Inverter

Manufacturer	Generic
Model	6250KVA-MV

(Original PVsyst database)

Unit Nom. Power	6874 kWac
Number of inverters	17 units
Total power	116858 kWac
Operating voltage	875-1300 V
Max. power (=>25°C)	7186 kWac
Pnom ratio (DC:AC)	1.14
Power sharing within this inverter	

Total inverter power

Total power	116858 kWac
Max. power	122162 kWac
Number of inverters	17 units
Pnom ratio	1.14

Array losses

Array Soiling Losses

Loss Fraction	2.0 %
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Thermal Loss factor

Module temperature according to irradiance	
Uc (const)	29.0 W/m²K
Uv (wind)	0.0 W/m²K/m/s

DC wiring losses

Global array res.	0.11 mΩ
Loss Fraction	1.5 % at STC

LID - Light Induced Degradation

Loss Fraction	2.0 %
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Module Quality Loss

Loss Fraction	-1.3 %
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Module mismatch losses

Loss Fraction	2.0 % at MPP
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Strings Mismatch loss

Loss Fraction	0.2 %
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Module average degradation

Year no	25
Loss factor	0.4 %/year

Mismatch due to degradation

Imp RMS dispersion	0.4 %/year
Vmp RMS dispersion	0.4 %/year

**Array losses****IAM loss factor**Incidence effect (IAM): Fresnel, AR coating, $n(\text{glass})=1.526$, $n(\text{AR})=1.290$

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.962	0.892	0.816	0.681	0.440	0.000

Spectral correction

FirstSolar model

Precipitable water estimated from relative humidity

Coefficient Set	C0	C1	C2	C3	C4	C5
Monocrystalline Si	0.85914	-0.02088	-0.0058853	0.12029	0.026814	-0.001781



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Horizon definition

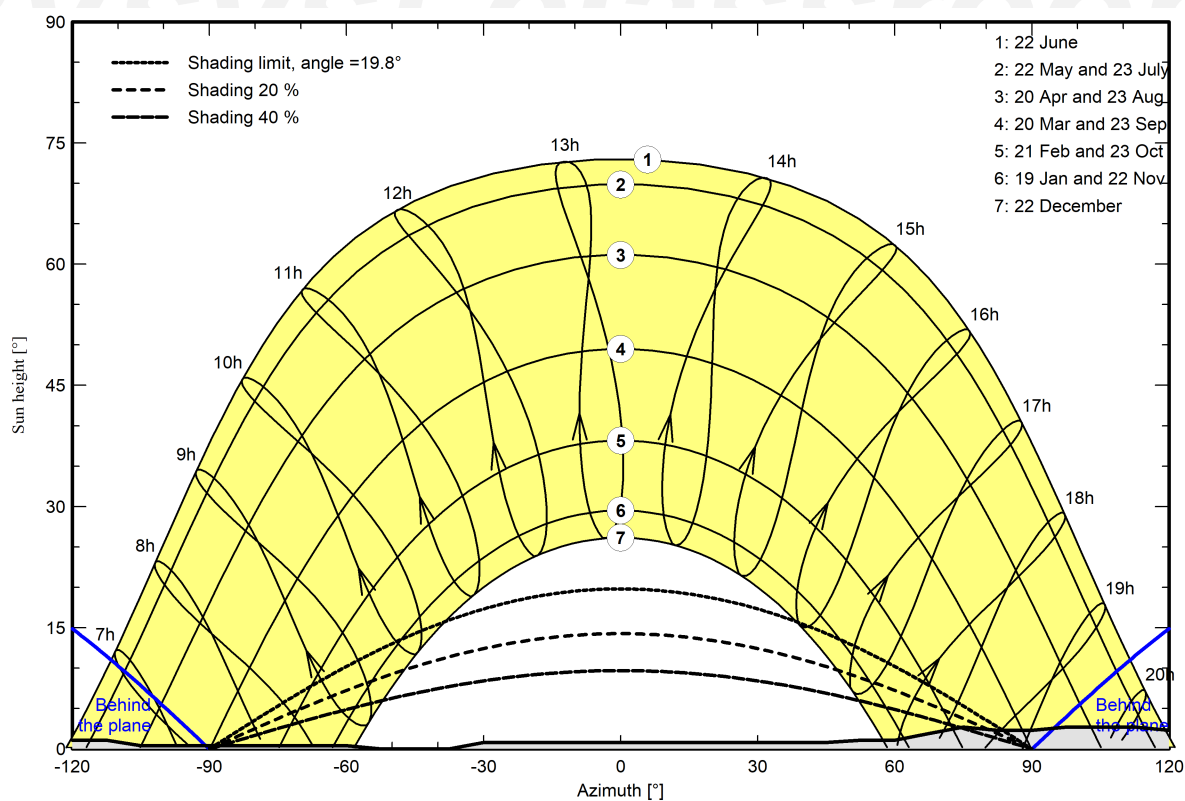
Horizon from PVGIS website API, Lat=40°25'13", Long=-3°35'27", Alt=664m

Average Height	1.2 °	Albedo Factor	0.96
Diffuse Factor	1.00	Albedo Fraction	100 %

Horizon profile

Azimuth [°]	-180	-173	-165	-143	-135	-113	-105	-60	-53	-38	-30	45	53	60
Height [°]	1.1	1.1	0.8	0.8	1.1	1.1	0.4	0.4	0.0	0.0	0.8	0.8	1.1	1.1
Azimuth [°]	68	75	83	90	98	113	120	128	135	158	165	173	180	
Height [°]	1.9	2.7	2.3	2.3	2.7	2.7	2.3	1.9	2.3	2.3	1.9	1.9	1.1	

Sun Paths (Height / Azimuth diagram)





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Main results

System Production

Produced Energy (Sim)	199298008 kWh/year	Specific production (Sim.)	1502 kWh/kWp/year	Perf. Ratio PR	71.46 %
Produced Energy (P50)	1290988 kWh/year	Specific production (P50)	1517 kWh/kWp/year		
Produced Energy (P75)	108312 kWh/year	Specific production (P75)	1485 kWh/kWp/year		
Produced Energy (P95)	1097016 kWh/year	Specific production (P95)	1440 kWh/kWp/year		

Economic evaluation

Investment

Global	292,420,787.50 GBP
Specific	2.20 GBP/Wp

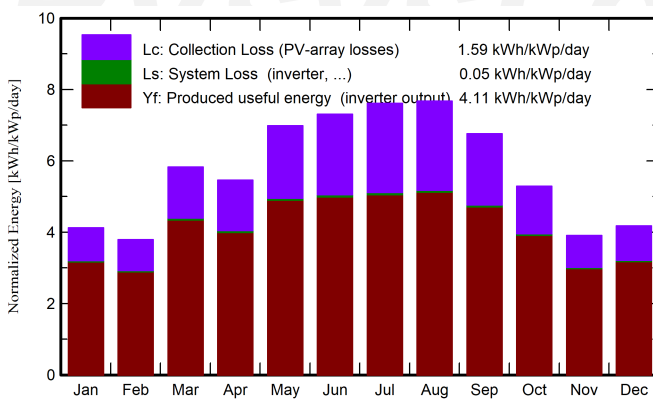
Yearly cost

Annuities	3,194,697.10 GBP/yr
Run. costs	7,577,800.80 GBP/yr
Payback period	13.0 years

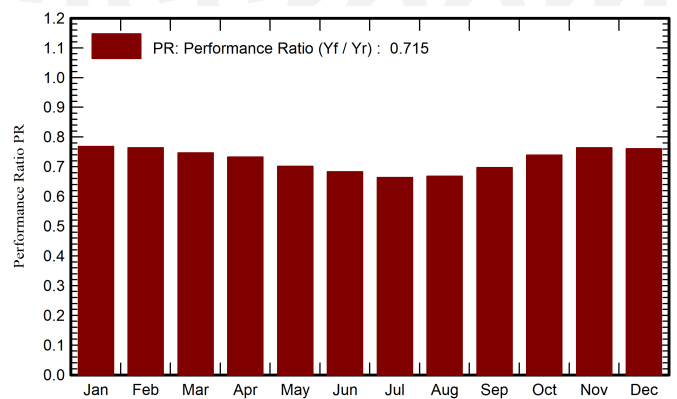
LCOE

Energy cost	0.12 GBP/kWh
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Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

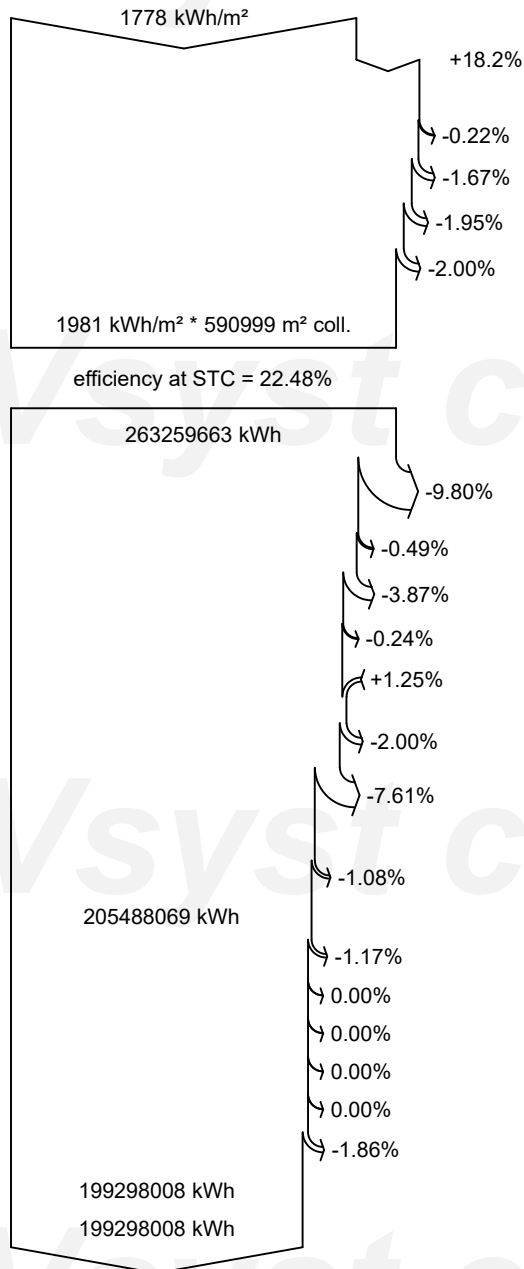
	GlobHor kWh/m ²	DiffHor kWh/m ²	T_Amb °C	GlobInc kWh/m ²	GlobEff kWh/m ²	EArray kWh	E_Grid kWh	PR ratio
January	75.2	23.27	5.77	127.8	119.7	13185826	13030649	0.768
February	77.1	37.12	6.28	106.1	99.6	10892483	10753532	0.764
March	142.0	44.83	9.58	180.6	171.3	18100269	17894367	0.747
April	151.7	67.03	11.94	163.8	154.3	16124331	15920218	0.732
May	215.3	63.56	19.22	216.6	204.4	20403694	20160320	0.702
June	226.9	66.81	23.62	219.3	206.7	20117743	19860267	0.683
July	239.5	62.66	27.81	236.1	222.9	21058611	20802762	0.664
August	221.9	50.72	27.04	237.8	225.4	21315055	21072614	0.668
September	166.8	44.76	22.92	202.9	192.8	18979986	18760160	0.697
October	117.7	42.39	16.64	164.0	155.2	16280757	16091766	0.739
November	74.0	27.90	7.67	117.3	110.1	12027226	11882069	0.763
December	69.5	19.77	5.63	129.5	119.1	13220775	13069284	0.761
Year	1777.8	550.82	15.40	2101.7	1981.5	201706757	199298008	0.715

Legends

GlobHor	Global horizontal irradiation	EArray	Effective energy at the output of the array
DiffHor	Horizontal diffuse irradiation	E_Grid	Energy injected into grid
T_Amb	Ambient Temperature	PR	Performance Ratio
GlobInc	Global incident in coll. plane		
GlobEff	Effective Global, corr. for IAM and shadings		



Loss diagram



Global horizontal irradiation

Global incident in coll. plane

Far Shadings / Horizon

Near Shadings: irradiance loss

IAM factor on global

Soiling loss factor

Effective irradiation on collectors

PV conversion

Array nominal energy (at STC effic.)

Module Degradation Loss (for year #25)

PV loss due to irradiance level

PV loss due to temperature

Spectral correction

Module quality loss

LID - Light induced degradation

Mismatch loss, modules and strings
(including 5.5% for degradation dispersion)

Ohmic wiring loss

Array virtual energy at MPP

Inverter Loss during operation (efficiency)

Inverter Loss over nominal inv. power

Inverter Loss due to max. input current

Inverter Loss over nominal inv. voltage

Inverter Loss due to power threshold

Inverter Loss due to voltage threshold

Available Energy at Inverter Output

Energy injected into grid



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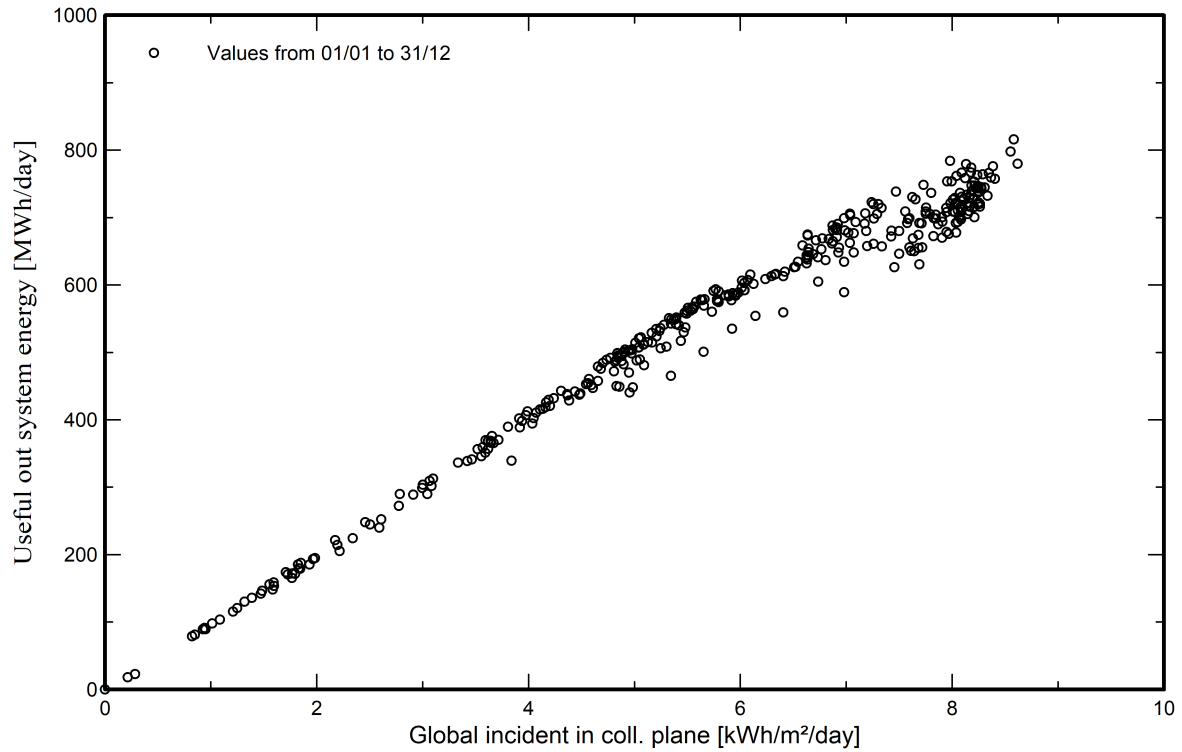
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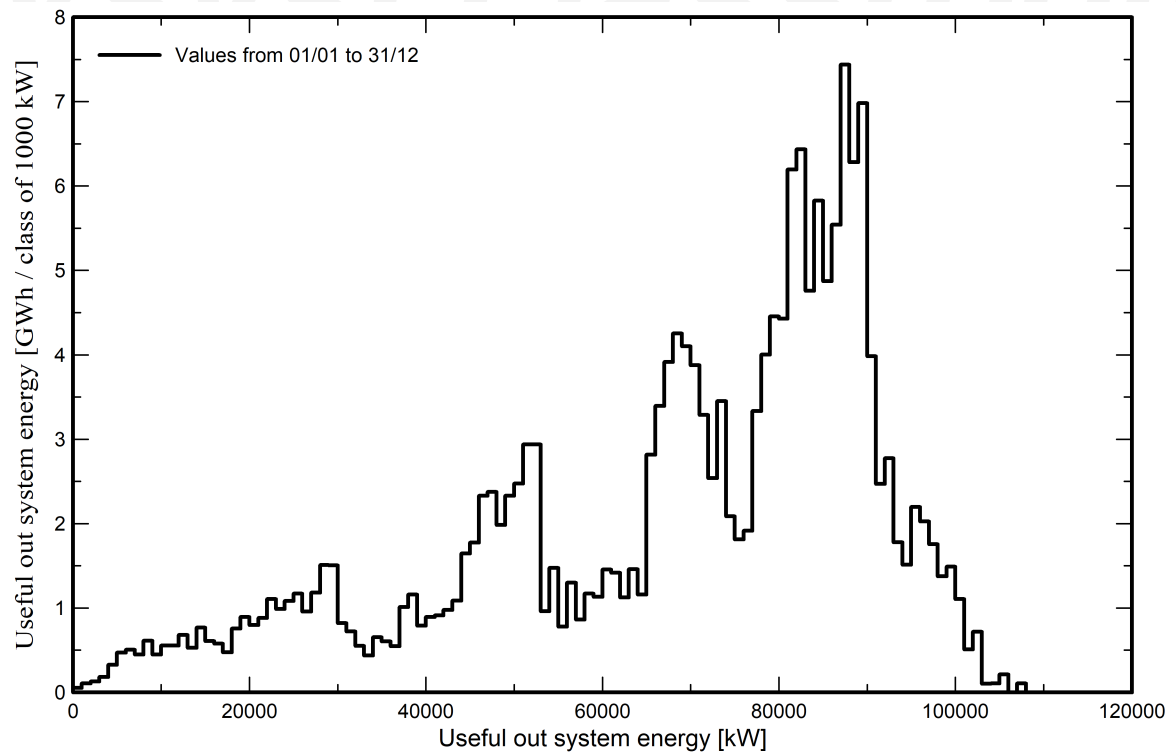
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Predef. graphs

Daily Input/Output diagram



System Output Power Distribution





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Aging Tool

Aging Parameters

Time span of simulation 25 years

Module average degradation

Loss factor 0.4 %/year

Mismatch due to degradation

Imp RMS dispersion 0.4 %/year

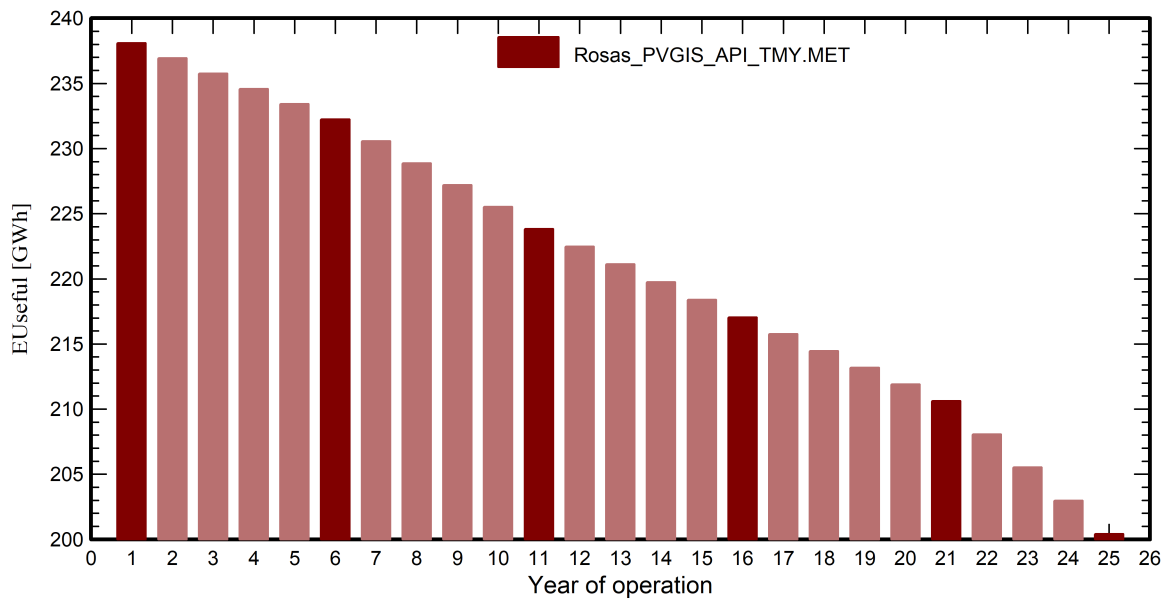
Vmp RMS dispersion 0.4 %/year

Weather data used in the simulation

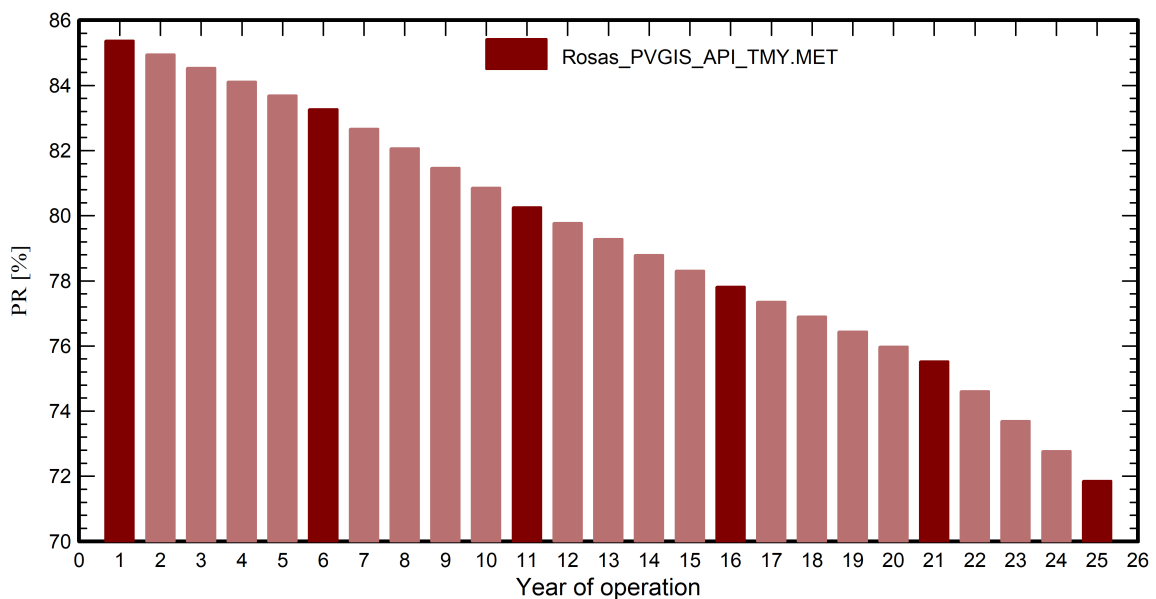
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Years reference year

Useful out system energy



Performance Ratio





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Aging Tool

Aging Parameters

Time span of simulation 25 years

Module average degradation

Loss factor 0.4 %/year

Mismatch due to degradation

Imp RMS dispersion 0.4 %/year

Vmp RMS dispersion 0.4 %/year

Weather data used in the simulation

Rosas PVGIS API TMY

Years reference year

	EUseful	PR	PR loss
Year	GWh	%	%
1	238.1	85.37	-0.25
2	236.9	84.95	-0.74
3	235.8	84.53	-1.23
4	234.6	84.11	-1.72
5	233.4	83.69	-2.21
6	232.2	83.27	-2.70
7	230.6	82.67	-3.41
8	228.9	82.07	-4.11
9	227.2	81.46	-4.81
10	225.5	80.86	-5.52
11	223.8	80.26	-6.22
12	222.5	79.77	-6.79
13	221.1	79.28	-7.36
14	219.7	78.80	-7.93
15	218.4	78.31	-8.50
16	217.0	77.82	-9.07
17	215.7	77.36	-9.61
18	214.5	76.90	-10.15
19	213.2	76.44	-10.68
20	211.9	75.98	-11.22
21	210.6	75.52	-11.76
22	208.1	74.61	-12.83
23	205.5	73.69	-13.90
24	203.0	72.78	-14.97
25	200.4	71.86	-16.03



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P50 - P90 evaluation

Weather data

Source PVGIS api TMY
Kind Monthly averages
TMY - Multi-year average
Year-to-year variability(Variance) 2.5 %

Specified Deviation

Climate change 1.0 %

Global variability (weather data + system)

Variability (Quadratic sum) 3.1 %

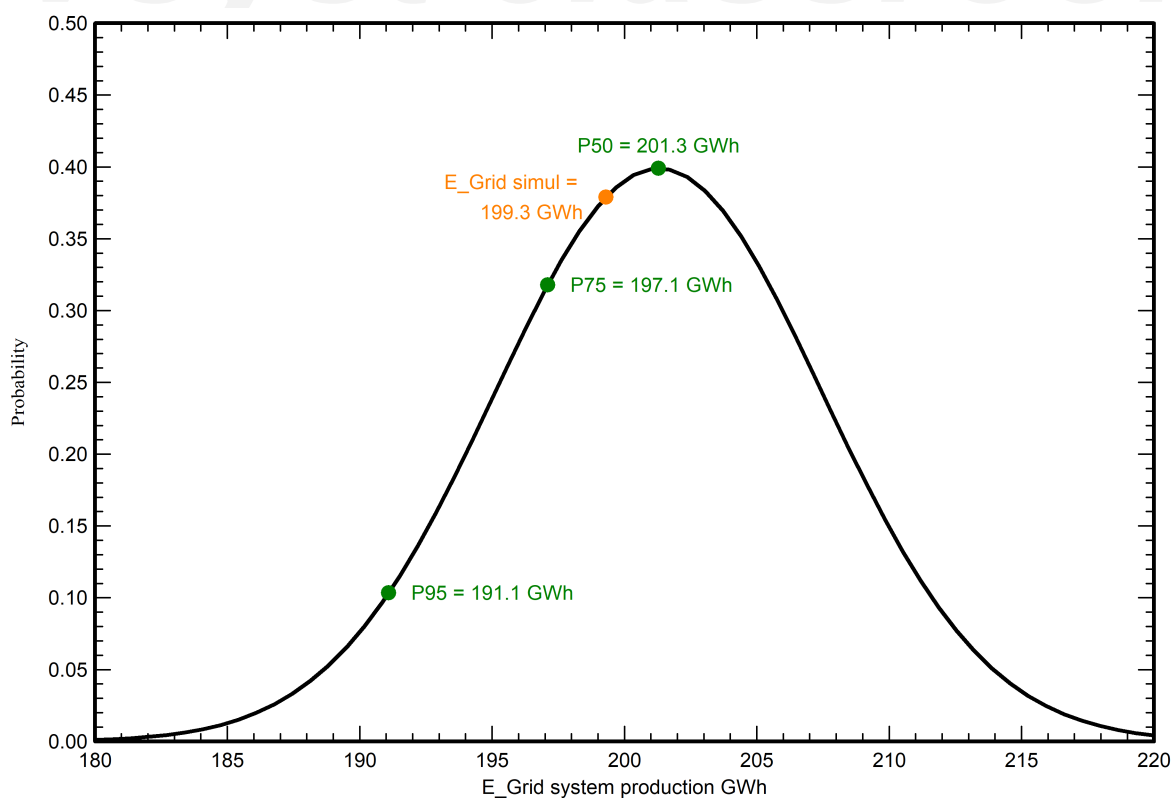
Simulation and parameters uncertainties

PV module modelling/parameters 1.0 %
Inverter efficiency uncertainty 0.5 %
Soiling and mismatch uncertainties 1.0 %
Degradation uncertainty 1.0 %

Annual production probability

Variability 6.2 GWh
P50 201.3 GWh
P75 197.1 GWh
P95 191.1 GWh

Probability distribution

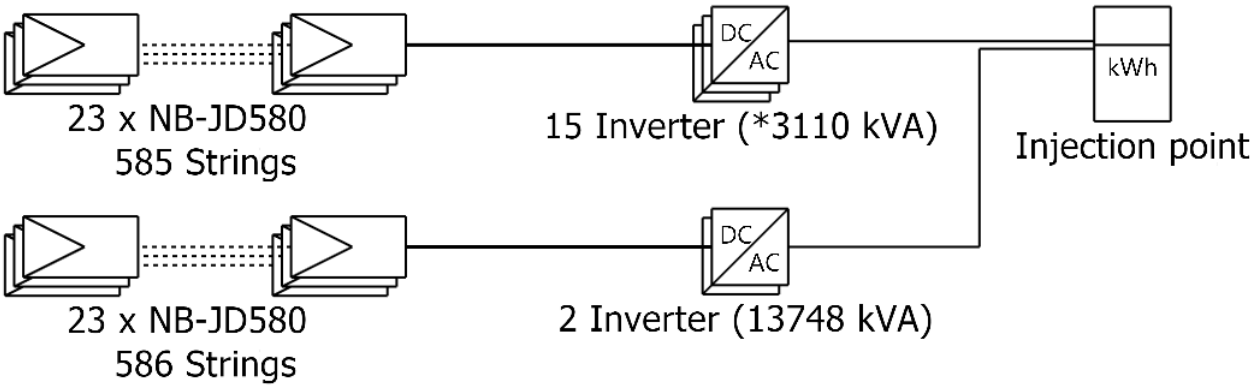




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Single-line diagram



PV module	NB-JD580
Inverter	6250KVA-MV
String	23 x NB-JD580

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Cost of the system

Installation costs

Item	Quantity units	Cost GBP	Total GBP
PV modules			
NB-JD580	228781	171.99	39,348,046.43
Supports for modules	228781	0.60	137,269.46
Inverters			
6250KVA-MV	17	50,000.00	850,000.06
Other components			
Wiring	150	58,491.46	8,773,719.69
Monitoring system, display screen	150	7,311.43	1,096,714.96
Measurement system, pyranometer	150	10,967.15	1,645,072.44
Studies and analysis			
Engineering	1	39,000.00	39,000.00
Installation			
Transport	1	8,067,063.19	8,067,063.19
Settings	150	694,586.14	104,187,921.30
Grid connection	150	54,835.75	8,225,362.21
Insurance			
Building insurance	1	182,785.83	182,785.83
Transport insurance	1	109,671.50	109,671.50
Land costs			
Land purchase	668083	138.05	92,230,341.34
Land preparation	150	58,491.46	8,773,719.69
Land taxes	1	0.00	10,283,683.06
Taxes			
VAT	1	0.00	8,470,416.35
		Total	292,420,787.50
		Depreciable asset	40,335,315.95

Operating costs

Item	Total GBP/year
Maintenance	
Provision for inverter replacement	34,000.00
Salaries	899,173.57
Repairs	187,887.01
Insurance	
Facilities insurance	268,410.02
Administrative, accounting	536,820.04
Total (OPEX)	1,926,290.64
Including inflation (10.00%)	7,577,800.80

System summary

Total installation cost	292,420,787.50 GBP
Operating costs (incl. inflation 10.00%/year)	7,577,800.80 GBP/year
Produced Energy	199298 MWh/year
Cost of produced energy (LCOE)	0.1211 GBP/kWh



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Financial analysis

Simulation period

Project lifetime 25 years Start year 2025

Income variation over time

Inflation 10.00 %/year
Production variation (aging) Aging tool results
Discount rate 8.00 %/year

Income dependent expenses

Income tax rate 47.00 %/year
Other income tax 0.00 %/year
Dividends 0.00 %/year

Depreciable assets

Asset	Depreciation method	Depreciation period (years)	Salvage value (GBP)	Depreciable (GBP)
PV modules				
NB-JD580	Straight-line	20	0.00	39,348,046.43
Supports for modules	Straight-line	20	0.00	137,269.46
Inverters				
6250KVA-MV	Straight-line	20	0.00	850,000.06
		Total	0.00	40,335,315.95

Financing

Own funds 146,210,393.73 GBP
Loan 1 - Interest-only bullet loan - 25 years 73,105,196.87 GBP Interest rate: 0.01%/year
Loan 2 - Interest-only bullet loan - 25 years 73,105,196.87 GBP Interest rate: 4.36%/year

Electricity sale

Feed-in tariff 0.38472 GBP/kWh
Duration of tariff warranty 20 years
Annual connection tax 0.00 GBP/kWh
Annual tariff variation 0.0 %/year
Feed-in tariff decrease after warranty 0.00 %

Return on investment

Payback period 12.3 years
Net present value (NPV) 227,753,950.21 GBP
Internal rate of return (IRR) 26.02 %
Return on investment (ROI) 77.9 %



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Financial analysis

Detailed economic results (kGBP)

Year	Electricity sale	Own funds	Loan principal	Loan interest	Run. costs	Deprec. allow.	Taxable income	Taxes	After-tax profit	Cumul. profit	% amorti.
0	0	146,210,394	0	0	0	0	0	0	0	-146,210,394	0.0%
1	76,674,131	0	0	3,194,697	1,926,291	2,016,766	69,536,378	32,682,097	38,871,046	-110,218,685	12.3%
2	76,674,131	0	0	3,194,697	2,118,920	2,016,766	69,343,749	32,591,562	38,768,953	-76,980,556	23.7%
3	76,674,131	0	0	3,194,697	2,330,812	2,016,766	69,131,857	32,491,973	38,656,650	-46,293,662	34.2%
4	76,674,131	0	0	3,194,697	2,563,893	2,016,766	68,898,775	32,382,424	38,533,117	-17,970,670	43.9%
5	76,674,131	0	0	3,194,697	2,820,282	2,016,766	68,642,386	32,261,921	38,397,230	8,161,839	52.8%
6	76,674,131	0	0	3,194,697	3,102,310	2,016,766	68,360,358	32,129,368	38,247,756	32,264,413	61.0%
7	76,674,131	0	0	3,194,697	3,412,541	2,016,766	68,050,127	31,983,560	38,083,333	54,485,672	68.6%
8	76,674,131	0	0	3,194,697	3,753,796	2,016,766	67,708,873	31,823,170	37,902,468	74,963,197	75.6%
9	76,674,131	0	0	3,194,697	4,129,175	2,016,766	67,333,493	31,646,742	37,703,517	93,824,342	82.1%
10	76,674,131	0	0	3,194,697	4,542,093	2,016,766	66,920,576	31,452,671	37,484,671	111,186,998	88.0%
11	76,674,131	0	0	3,194,697	4,996,302	2,016,766	66,466,366	31,239,192	37,243,940	127,160,285	93.5%
12	76,674,131	0	0	3,194,697	5,495,932	2,016,766	65,966,736	31,004,366	36,979,136	141,845,209	98.5%
13	76,674,131	0	0	3,194,697	6,045,525	2,016,766	65,417,143	30,746,057	36,687,852	155,335,256	103.1%
14	76,674,131	0	0	3,194,697	6,650,078	2,016,766	64,812,591	30,461,918	36,367,439	167,716,952	107.4%
15	76,674,131	0	0	3,194,697	7,315,086	2,016,766	64,147,583	30,149,364	36,014,985	179,070,377	111.2%
16	76,674,131	0	0	3,194,697	8,046,594	2,016,766	63,416,074	29,805,555	35,627,285	189,469,642	114.8%
17	76,674,131	0	0	3,194,697	8,851,253	2,016,766	62,611,415	29,427,365	35,200,816	198,983,329	118.0%
18	76,674,131	0	0	3,194,697	9,736,379	2,016,766	61,726,289	29,011,356	34,731,699	207,674,904	121.0%
19	76,674,131	0	0	3,194,697	10,710,017	2,016,766	60,752,652	28,553,746	34,215,671	215,603,087	123.7%
20	76,674,131	0	0	3,194,697	11,781,018	2,016,766	59,681,650	28,050,375	33,648,040	222,822,214	126.2%
21	76,674,131	0	0	3,194,697	12,959,120	0	60,520,314	28,444,548	32,075,766	229,194,249	128.4%
22	76,674,131	0	0	3,194,697	14,255,032	0	59,224,402	27,835,469	31,388,933	234,967,946	130.4%
23	76,674,131	0	0	3,194,697	15,680,535	0	57,798,899	27,165,482	30,633,416	240,185,285	132.1%
24	76,674,131	0	0	3,194,697	17,248,589	0	56,230,845	26,428,497	29,802,348	244,885,095	133.7%
25	76,674,131	0	146,210,394	3,194,697	18,973,448	0	54,505,986	25,617,814	-117,322,221	227,753,950	177.9%
Total	1,916,853,280	146,210,394	146,210,394	79,867,428	189,445,020	40,335,316	1,607,205,516	755,386,593	745,943,846	227,753,950	177.9%



Project: Madrid_Fixed_Angle_PV_Syst

Variant: Madrid_150MW - Simulation for year no 25 - Simulation for year no 25 - Simulatio

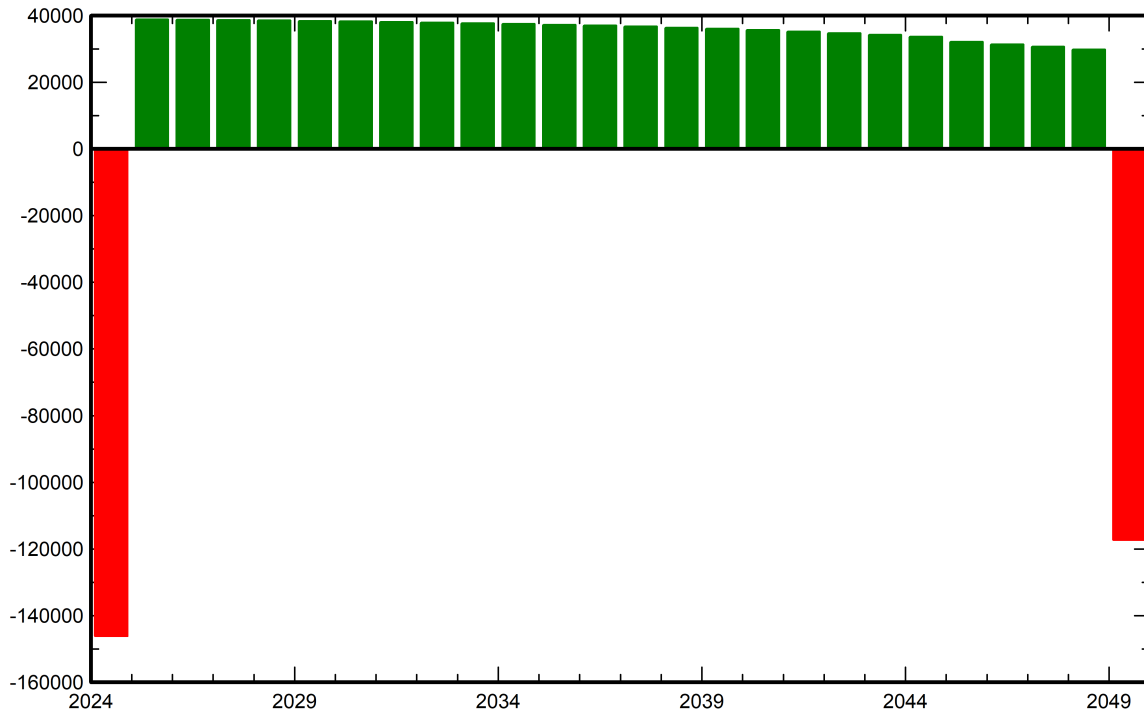
PVsyst V7.4.6

VCO, Simulation date:
31/03/24 12:16
with V7.4.6

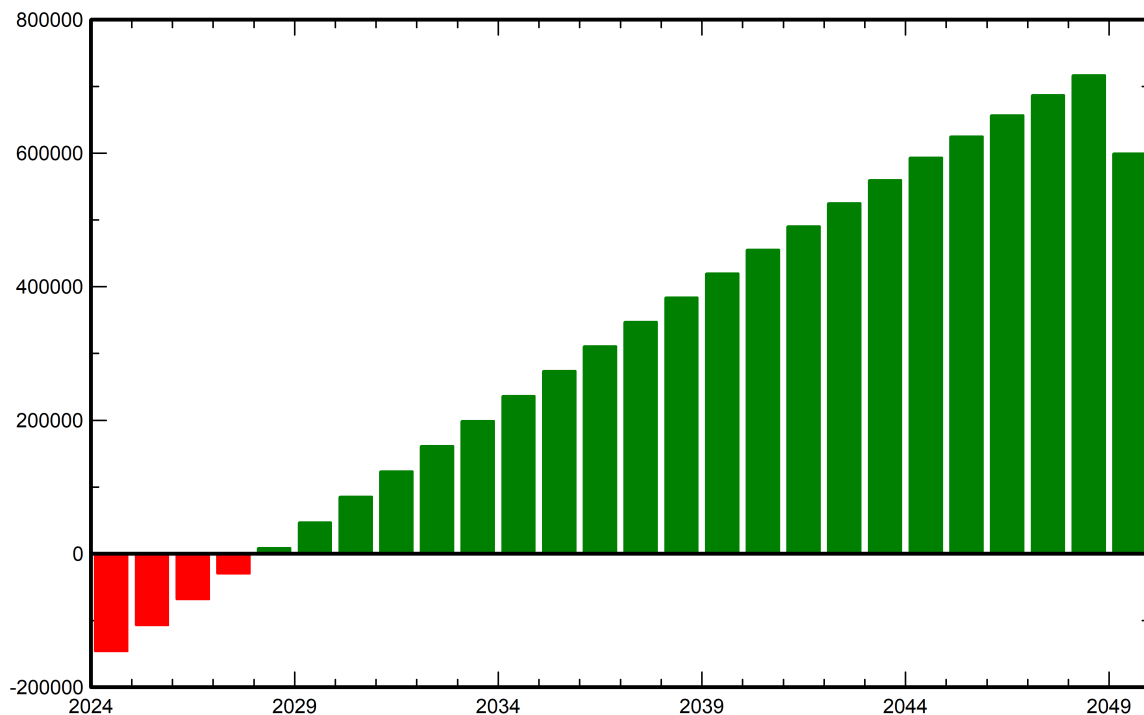
University of Edinburgh (United kingdom)

Financial analysis

Yearly net profit (kGBP)



Cumulative cashflow (kGBP)





PVsyst V7.4.6

VCO, Simulation date:
31/03/24 12:16
with V7.4.6

University of Edinburgh (United kingdom)

CO₂ Emission BalanceTotal: 1257226.9 tCO₂

Generated emissions

Total: 231648.68 tCO₂

Source: Detailed calculation from table below

Replaced Emissions

Total: 1715955.8 tCO₂

System production: 199298.01 MWh/yr

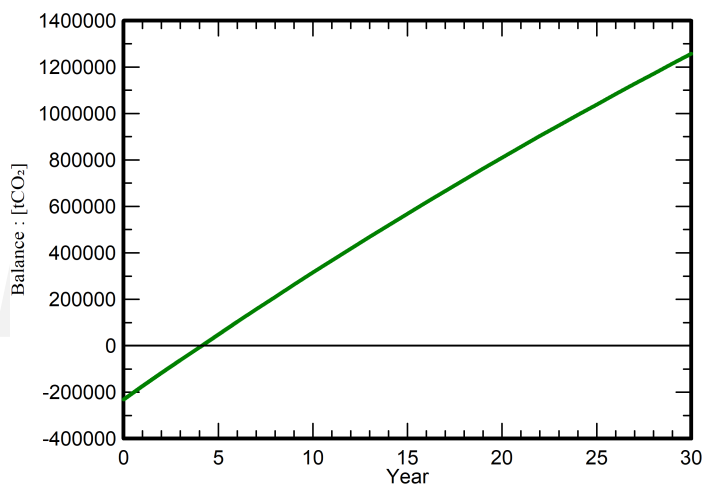
Grid Lifecycle Emissions: 287 gCO₂/kWh

Source: IEA List

Country: Spain

Lifetime: 30 years

Annual degradation: 1.0 %

Saved CO₂ Emission vs. Time

System Lifecycle Emissions Details

Item	LCE	Quantity	Subtotal
			[kgCO ₂]
Modules	1713 kgCO ₂ /kWp	132693 kWp	227265921
Supports	1.91 kgCO ₂ /kg	2287810 kg	4379532
Inverters	190 kgCO ₂ /units	17.0 units	3224